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ORIGINAL ARTICLE

SAMCEP Society consensus on the treatment of upper facial lines with botulinum neurotoxin type A: A tailored approach

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Abstract

Background: The safety and efficacy of botulinum neurotoxin type A (BoNTA) treatments are well established, but injection techniques, target muscles, and toxin doses continue to evolve, with each refinement producing improvements in treatment outcomes. The recommendations in this consensus move away from standard templates and illustrate how to tailor treatments to individual patterns and strengths of muscle activity, and patient preferences.

Methods: Seventeen experts in the fields of plastic surgery, dermatology, ophthalmology, otorhinolaryngology, and neurology convened in 2022 to develop consensus-based recommendations for the use of botulinum toxin A for the treatment of horizontal forehead lines, glabellar frown lines, and crow's feet lines that reflect current clinical practice. The focus was on how to tailor injections to individual patients to optimize treatment outcomes.

Results: For each upper face indication, consensus members describe how to perform a dynamic assessment to optimize the dose and injection technique for each patient. A tailored treatment protocol is presented for commonly observed patterns of dynamic lines. Units of Inco are defined and the precise location of injection points, illustrated with the use of anatomical images.

Conclusion: This consensus provides up-to-date recommendations on the tailored treatment of upper facial lines based on the latest research and collective clinical experience of the expert injectors. Optimal outcomes require thorough patient evaluation, both at rest and during animation, using both visual and tactile cues; detailed understanding of facial muscular anatomy and how opposing muscles interact; and use of a BoNTA with high precision to target identified zones of excess muscle activity.

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KEYWORDS
aesthetics, consensus guidelines, crow's feet, glabellar frown lines

1 | INTRODUCTION

Demand for facial injectable treatments, both botulinum neurotoxin type A (BoNTA) and dermal fillers, continues to grow with the greater public awareness and acceptance of facial aesthetic enhancement. This trend is illustrated by data from the International Society of Aesthetic Plastic Surgery (ISAPS), which showed that dermal filler and BoNTA treatments increased by 50.6% and 35.6%, respectively, over the period 2015 to 2019.[1] Worldwide and in most European countries, BoNTA injections are the most performed nonsurgical aesthetic procedure in both women and men in all age groups.[1] In France, the popularity of BoNTA injections also ranks highly but unlike other European countries is second to hyaluronic acid injections in terms of the number of procedures performed.[1]

Five BoNTAs are currently available on the French market. Onabotulinumtoxin A (Ona; Vistabel/Botox, Allergan Inc., Irvine, CA, USA) received marketing authorization from the Agence Française de Sécurité Sanitaire des Produits de Santé (AFSSAPS) in 2003 for glabellar frown lines, followed by abobotulinumtoxin A (Abo; Azzalure/Dysport, Galderma, Paris, France) in 2009, and incobotulinumtoxin A (Inco; Bocouture/Xeomin, Merz Aesthetics GmbH, Frankfurt Germany) in 2010. All three toxins subsequently received marketing authorizations for the treatment of lateral canthal lines (crow's feet), and Inco was the first BoNTA to additionally receive marketing authorization for the treatment of all three types of upper facial lines (glabellar, crow's feet, and horizontal forehead lines). The most recent addition to the market is letibotulinumtoxin (Letybo, Croma-Pharma) approved for glabellar lines in February 2022. The BoNTA market is set to further expand in the near future with the recent approval of a liquid form of abobotulinumtoxin A for glabellar lines (Alluzience, Galderma).

Although all BoNTAs share a common mechanism of action resulting in a presynaptic block of acetylcholine release, their manufacturing process, structure, pharmacological properties, and stabilizing excipients are distinct, conferring differences in their potency and immunogenicity, with implications for differing clinical efficacy and safety.[2,3] The three reference BoNTA formulations also differ in their content of nontoxin complexing proteins, Inco being the only preparation to contain only the pure neurotoxin. The complexing proteins associated with Ona and Abo do not play a role in any of the steps involved in the blockade of neurotransmitter release, neither are they required for the stability of the toxin complex or for limiting the spread of the neurotoxin.[4,5] While the complexing proteins have no effect on the treatment response, they have the potential to act as adjuvants and generate a heightened immune response. As a pure neurotoxin, Inco therefore has the lowest immunogenic potential of the three formulations.[6,7]

To date, there are no internationally accepted standardized tests for BoNTA comparisons. Each product has its own proprietary assay method for determining potency. As a result of variations of biological activity and differences in biological assays, regulatory agencies in most countries worldwide require a statement of unit noninterchangeability among BoNTA products. The final treatment result reflects the quantity of picograms of active neurotoxin injected in every single point.

The use of BoNTA for aesthetic purposes is evolving rapidly. Approved indications are still restricted to the treatment of horizontal forehead lines, glabellar lines, and crow's feet, but the products are also widely used off-label for other facial indications including eyebrow shaping, infraorbital eye and upper eyelid opening, perioral lines, transverse nasal rhytids (Bunny Lines), excessive gingival display (Gummy Smile), labial lip flip, correction of labial inversion while smiling, cobblestone chin, lifting of oral commissures, square jaw, and softening platysmal bands.[8] The experience gained has led to the refinement of techniques, targeted to take into account differences in patients' anatomy, muscle strength, and density, which can change as a person ages. Patient preferences also vary and change. BoNTA injections are now increasingly requested at earlier ages with the recognition that they are useful for both the prevention and treatment of facial lines and stabilization of the face, mainly the position of the eyebrow tails and the corner of the mouth, rather than just treatment alone.[9] Treatments must also be precise and allow some facial movement so that patients look natural and the expressions defining their personality are maintained. This has led to the development of several tailored treatment protocols to accommodate individual anatomic variations and preferences.[10,11] As our experience with injection of BoNTAs increases, so too do the number of off-label indications, for example, the inferior third of the face.[12]

In France, neurotoxin injections can only be prescribed and injected by authorized health professionals. Physician injectors have received training, dealing solely with facial aesthetics, in terms of anatomy, safety, balancing facial features and achieving a natural-looking aesthetic result. This knowledge determines which muscle to target, at what depth, and at what dose, how this changes with aging of the face, and how to practice preventive treatment. Several international consensus recommendations and guidelines have been published in the last few years.[8,10,13–16] The aim of the current consensus was to provide detailed and tailored injection protocols for the three upper facial indications approved in France, combining the experts' significant injection experience with the latest anatomical knowledge and injection techniques. This information is intended to assist injectors with less experience to develop the skills for optimal and safe patient outcomes.

2 | WHY WAS THIS EXPERT CONSENSUS REQUIRED?

Injection techniques and BoNTA uses have significantly evolved since the reference BoNTA products were originally approved, and it is now recognized that the standardized treatment protocols required for regulatory approval are not optimal for all patients.[17] In addition to gender and racial differences in the facial skeleton, muscle mass, and skin thickness, individuals also vary in terms of muscle shape, size, strength, and orientation, all of which influence how they will respond to treatment. These variations may also be observed within an individual, with one study reporting that at least 20% of individuals have significant asymmetry from one side of the face to the other.[18] In clinical practice, a one-size fits all approach to BoNTA treatment is therefore rarely appropriate and can lead to suboptimal or excessive doses and unsatisfied patients.

The alternative, described in this consensus document, is to base treatment on standard injection protocols but then adapt to the individual patient following a thorough pretreatment evaluation. With this approach, the insertion points and doses of BoNTA delivered are adapted based on the size, shape, and strength of the muscles to be targeted, and on whether the patient is seeking preventive wrinkle avoidance or treatment of established lines.

3 | CONSENSUS METHODOLOGY

Seventeen French experts in the fields of plastic surgery, dermatology, ophthalmology, otorhinolaryngology, and neurology convened five times between February and November 2022 to develop independent, consensus-based recommendations for the use of Inco for the treatment of upper facial lines: horizontal forehead lines, glabellar frown lines, and crow's feet lines. Experts were selected to be part of the panel based on their medical expertise and experience and their familiarity with the use of BoNTA (mean duration of experience with BoNTA products was 18 years).

During these meetings, the group developed guidance on the optimal treatment of upper face BoNTA indications based on the severity of the defect, and anatomy, gender and age of the patient. The focus was on how to tailor injections to individual patients to optimize treatment outcomes. The following text summarizes the recommendations of the group. Nonreferenced statements represent the opinions of the panel and are not intended to be statements of fact.

4 | CONSENSUS RECOMMENDATIONS

4.1 | Patient evaluation and classification

BoNTA is generally considered to have a wide safety margin, and in experienced hands is safe and effective for a variety of facial aesthetic indications.[19] Patients with dynamic rhytids show the most dramatic improvements from BoNTA injections and are ideal candidates for treatment. Deep static lines due to loss of skin elasticity, sun damage, and aging are not responsive to BoNTA injection.

A careful patient evaluation and setting realistic expectations at the time of consultation are essential for achieving patient satisfaction and optimal outcomes. Delicate and varied facial expressions are accomplished by the activity and interaction of numerous facial muscles. There can be significant variations in muscle size, shape, and strength between individuals, and these together with variations in muscle interactions and dominance influence the pattern of expression lines observed.

When evaluating a patient, experts recommended that physicians look for specific clinical findings that may influence treatment results such as asymmetry, pre-existing brow ptosis, lid ptosis, pattern of frontalis muscle motion, brow shape, and malar edema. It is important to palpate the forehead to determine an intermediate frontalis adherence and distinguish the region of frontalis muscle involved in eyebrow position. Physicians should also evaluate the Orbital Retaining Ligament with pressure over the eyebrow and eyes closed. Evaluations should be performed with the patient at rest and then in animation, including smiling, squinting, frowning, and elevating the eyebrows to assess the mass, size, thickness, and depth of the target muscle below the skin. This should be repeated throughout the procedure as observing muscle action is essential when determining injection points and deciding on dose.

Prior to treatment, physicians should discuss the risks, benefits, and alternatives of BoNTA treatment with the patient and also provide this information in written format and ensure that the patient provides informed and signed consent.

4.2 | Choice of botulinum neurotoxin and reconstitution

The focus of this consensus is based on treatment with Inco. While the duration of action of all BoNTAs varies from one patient to another, it is generally agreed that the available toxins have a similar duration of effect. Inco is unique in that it consists of only the pure neurotoxin and thus, in theory, may have a lower risk of neutralizing antibody formation (although the risks of this occurring in aesthetic use with current formulations of Ona and Abo are also very low). An added advantage of Inco is that unopened vials do not require refrigeration and can be stored at room temperature.

For aesthetic use, Inco is supplied as a powder in vials containing 50 or 100 U. It should be reconstituted with sterile, preservative-free 0.9% saline, after which it should be stored in a refrigerator at 2–8°C (36–46°F) and used within 24 h. Inco is reconstituted by injecting the saline directly into the toxin vial with a sterile needle and gently rotating and inverting so that the liquid reaches all surfaces of the vial and rubber cap. This ensures complete reconstitution and no loss of product.[20,21] When correctly reconstituted, Inco is a clear, colorless solution free of particulate matter. For the approved Inco aesthetic indications, the recommended injection volumes are

1.25 mL for the 50 U vial and 2.5 mL for the 100 U vial to give an Inco dose of 4 U/0.1 mL. Improper reconstitution can reduce or diminish the efficacy of BoNTA preparations, which could result in the inaccuracy of actual units injected and therefore suboptimal results.[20,21]

4.3 | Treatment intervals

In therapeutic indications, including blepharospasm, cervical dystonia, and poststroke spasticity, randomized, placebo-controlled trials have investigated flexible Inco injection intervals based on the individual patient's clinical need.[22–25] Results showed that the frequency of adverse events was similar whether intervals were as short as 6 weeks or ≥ 12 weeks, with no cases of newly formed neutralizing antibodies even at the high doses used in these therapeutic indications. In aesthetic indications, the recommended interval between two Inco injection sessions is ≥12 weeks.[26] The frequency of treatment-related adverse events is low with repeat injections at intervals of 12 weeks or more and appears to decrease further with subsequent treatments.[27,28] No studies have investigated repeat injections at intervals shorter than this in aesthetic indications. However, the specific pharmacological and immunological data related to a pure toxin without complexing proteins that are unique to Inco indicate that shorter injection intervals could also be applied in the aesthetic field. Experts agreed that injection intervals could be shortened or lengthened depending on the needs of the individual patient.

4.4 | Horizontal forehead lines

4.4.1 | Relevant anatomy

The frontalis is a large, paired, fan-shaped muscle with vertically-oriented fibers that originate in the galea aponeurotica, and inserts into the skin above the eyebrows. The frontalis has no direct bony attachment and is very mobile, and as the only levator of the brow, plays an important role in conveying emotions and nonverbal communication. It has a double role, to elevate the eyebrows and lower the hairline. These actions are split on a horizontal line, which is generally located under the main horizontal wrinkle of the forehead. In practice, it is difficult to establish a standard injection scheme for the horizontal forehead lines produced by contraction of the frontalis. There are many anatomic variations in the size of the frontal skeleton and therefore in the extent of muscle coverage, as well as in the shape and strength of the frontalis muscle. Four distinct patterns have been described, which reflect the shape and strength of the underlying frontalis (Figure 1): (1) full, straight lines that run across the whole forehead (45%); (2) V-shaped lines with a central depression and lateral elevation (30%); (3) short central horizontal lines over the middle of the forehead but few or no lines laterally (10%); and (4) lateral straight lines as two columns formed on the lateral aspect of the forehead with no central lines (15%).[11,29]

4.4.2 | Patient evaluation

Prior to any treatment, the patient should be asked to forcefully elevate their brows and the injector should note the degree of medial, central, and lateral brow elevation, as well as the shape of the brow arch, so that only areas with active muscle are injected. The patient should then be asked to relax their forehead, close their eyes, and the degree of brow ptosis assessed, with and without a hand on the forehead to test the deep ligaments. The examination should also determine any existing facial asymmetry and any compensated brow ptosis, where the patient subconsciously elevates the brow to compensate for the descent of the eyebrow from its normal anatomic position. This can be detected by determining whether there are any permanent contractions of the frontalis in the supraorbital region.

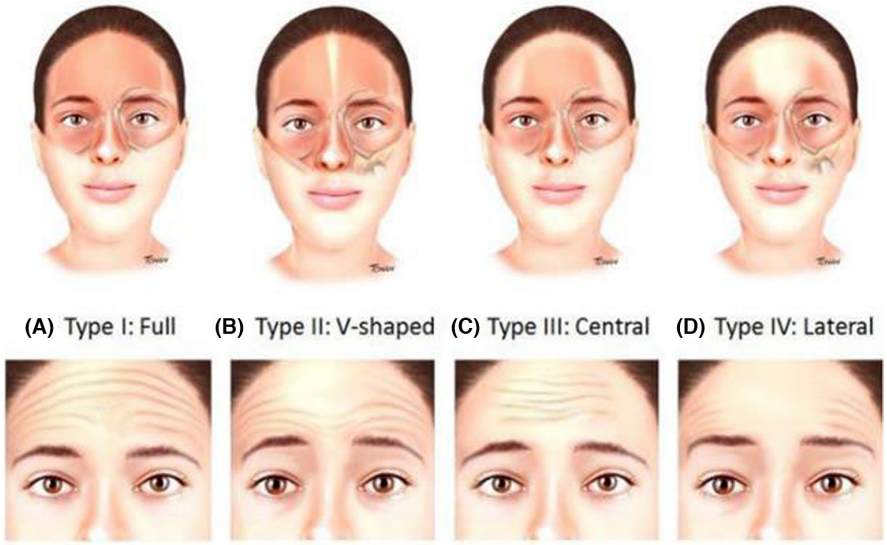


FIGURE 1 Variations in frontalis anatomy and corresponding forehead line patterns.[11]

4.4.3 | Standard injection protocol

The classic injection protocol for horizontal forehead lines recommends a total dose range of 10–20 U injected into the frontalis muscle in five horizontally aligned injection sites at least 2 cm above the orbital rim (Figure 2). Per injection point, 2 U (0.05 mL), 3 U (0.075 mL), or 4 U (0.1 mL) are applied, respectively. While this provides a basis for the treatment strategy, a personalized approach adapted to each patient (Table 1) makes it possible to optimize treatment and avoid the need for touch-ups because of insufficient results.

4.4.4 | Tailored injection protocol

To tailor BoNTA treatment of the frontalis to the individual, the experts recommended the One21 technique that has been designed for use with Inco, due to its high precision and low spread.[11,30] This consists of a 3-step protocol: (1) assessment to identify areas of hyperactivity, (2) bespoke to identify individuals desired results, and (3) customization with dosing and injecting. The forehead grid of 21 squares, representing 21 potential injection sites, is produced by drawing seven vertical lines (inner and outer canthus, mid-pupillary line, and mid-facial line) and three horizontal lines corresponding to the lowest and highest horizontal forehead lines (Figure 3). The physician then palpates the muscle force at each of the 21 points to determine where to place toxin (no muscle movement, no injection). Inco dosing will depend on the strength of the muscle. For a female, suggested doses are as follows: strong segment of frontalis (2 U), moderately strong segment of frontalis (1 U), weak segment of frontalis (0.5 U), and no muscle force palpated (no Inco required). For a male patient, the doses can be replaced by 3 U, 2 U, 1 U, and 0 U, respectively. Each of the 21 potential injection sites can be marked in a different color to indicate the dosage required, to assist in application. When using Abo, 2–4 U per point are normally required.

To avoid brow ptosis, the muscle fibers of the frontalis must remain fully functional 2 cm above the orbital rim, and all injections are performed above this level. Using this technique, injection points for individualized eyebrow reshaping and positioning are also easily determined, e.g., to arch and laterally raise the eyebrow, injections are performed medially to the mid-pupillary line.

4.4.5 | Mephisto management

The so-called "Mephisto look," in which the eyebrow tail is elevated, occurs in some patients when the lateral fibers of the frontalis are not sufficiently paralyzed compared with the medial fibers. It is more common when treatment of the forehead is restricted to the area between the mid-pupillary lines. This unaesthetic result can be prevented by performing a blocking test of the medial part of the forehead while asking the patient to raise their eyebrows. Expert recommendation was for patients with strong lateral frontalis fibers to receive 1 U intramuscularly in the lateral frontalis.[10] In some patients, an injection in the frontalis lateral to the mid-pupillary line can avoid the problem.

4.4.6 | Management of mild asymmetries

A cadaver study has reported significant frontalis muscle asymmetry in 20% of specimens.[18] Any facial asymmetries noted during the evaluation should be pointed out to the patient prior to treatment. There should be discussion on whether the asymmetry should remain or be corrected with BoNTA treatment.

4.4.7 | Management of supraciliary wrinkles

The treatment of supraciliary wrinkles in the lower lateral forehead is associated with a high risk of eyelid ptosis. Experts described two options for the management of these lines. The first respects the classic injection scheme with injections performed 2 cm above the orbital rim. A follow-up visit is then scheduled 2 to 4 weeks after the initial injection to determine whether a re-touch is necessary. The second is to perform a very superficial injection of the supraciliary lines with 1 U of Inco (1 U Abo). The objective is to avoid deep injection that could lead to diffusion of toxin to the levator palpebral muscle. The second option is preferred by experienced injectors. Inco is particularly useful in situations such as this where precision and limited diffusion are essential.

4.5 | Glabellar frown lines

4.5.1 | Relevant anatomy

Glabellar frown lines are associated with the expression of negative emotions and contribute to a person appearing older than they are. When treating these lines, the main BoNTA targets are the medial brow depressors: corrugator supercilii and the procerus (Figure 4).

The paired corrugator supercilii originate from the medial part of the superciliary arch, cross through the frontalis muscle, and insert into the skin laterally over the middle of the orbital arch.[31,32] On contraction, they pull the skin of the brow medially and lightly inferiorly to produce the vertical glabellar lines (Figure 4) and can be very powerful, especially in men. These muscles are thicker at the medial canthus than the mid-pupillary line. The corrugator dimple is the cutaneous end of the corrugator and is easy to find when a patient frowns.

The corrugators lie under the frontalis medially and on top of the frontalis laterally. Experts highlighted the importance of understanding the varying depth of the corrugator to avoid unintentional injection of the frontalis. For example, too superficial placement of BoNTA medially could target the frontalis muscle and cause the brows to drop. BoNTA injections of the corrugator should be made deep medially at its origin on bone, and very superficial laterally

TABLE 1 Injection protocol for horizontal forehead lines.

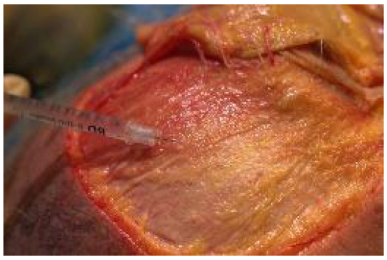
Cadaver dissection of left frontal muscle (the needle is in the muscle).		
	*Injection sites*	Location and number of injection sites determined using ONE21 approach based on areas of hyperactivity
	Dosage	
	Total	Dosing dependent on muscle strength
	Per injection site	0.5–2 U (women), 1–3 U (men), 2–4 U Abo
	Injection technique	Superficial intramuscular papula injection



FIGURE 2 Standard injection protocol for horizontal forehead lines

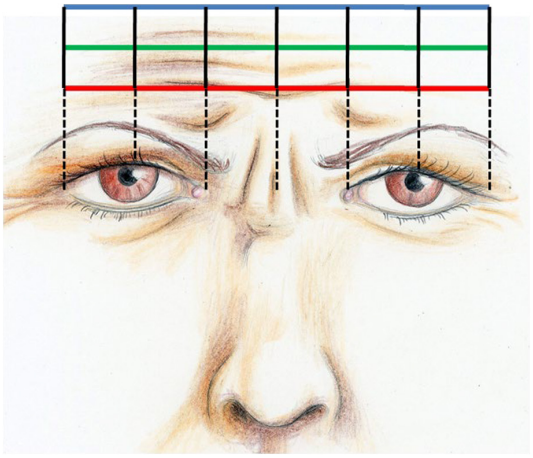


FIGURE 3 Tailored injection protocol for horizontal forehead lines: the ONE21 approach.[11] Red = inferior limit line. Green = non-mobile forehead line. Blue = superior limit line.

at the point of skin insertion. Moreover, the end of the corrugator overlaps the supraorbital pedicle, so it is important to stay superficial to avoid bleeding and an active diffusion of toxin to the orbital septum.

The procerus is a superficial pyramid-shaped muscle that lies over the frontalis. Its fibers originate at the midline of the nasal bone and run superiorly to merge with the frontalis muscle along the medial canthus and insert into the skin of the lower forehead between the eyebrows. The action of the procerus forms a horizontal line below the glabella by pulling down the medial side of the eyebrow. The strength of the procerus is very variable.

4.5.2 | Patient evaluation

Both the corrugator and procerus muscles have their medial origin in contact with bone and their terminal attachment laterally under the skin. The precise location of injection can only be determined by asking the patient to contract the muscles. For the corrugator, the patient should push their eyebrows together. The muscle can be felt during this animation, and the medial area is injected deeply where it originates on bone. The lateral injection point is determined by observing where the skin puckers and injecting just medial to this point.

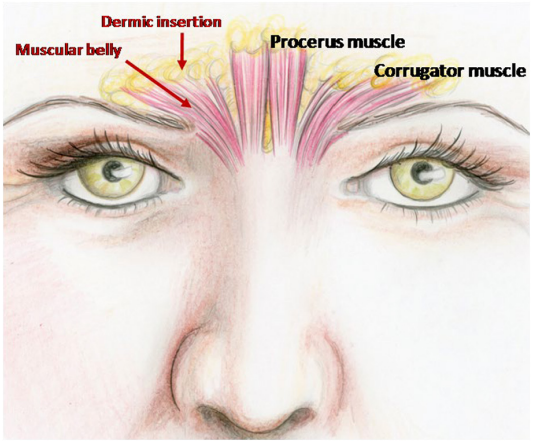


FIGURE 4 Facial muscles responsible for glabellar frown lines. The paired corrugator supercilii pull the skin of the brow medially and inferiorly to produce the vertical glabellar lines.

To determine the procerus insertion point the patient is again asked to bring their eyebrows together. The insertion point is just above the resulting horizontal rhytid. Injection is deep and the dose depends on the size and strength of the muscle.

4.5.3 | Standard injection protocol

Glabellar contraction patterns vary depending on the shape, orientation, strength, and insertion sites of the muscles involved and can be used as a tool to optimize BoNTA treatment. For the glabellar region, five contraction patterns have been identified (Figure 5).[33] The "U" pattern is the most commonly observed and is best treated with the classic five injection-site models where a dose of 4 U (0.1 mL) is injected into each of the five injection sites: two injections in each corrugator muscle and one injection in the procerus muscle that corresponds to a standard dose of 20 U (Table 2; Figure 6). For Abo, 10 U per point is the on-label suggestion.[34] The dose may be increased by the physician to up to 30 U if required by the individual needs of the patient.

4.5.4 | Tailored injection protocol

Treatment protocols for other glabellar contraction patterns may require adjustment depending on the muscles involved and their relative dominance.[35] Thus, hypertonic muscles are treated with higher doses or a greater number of injection sites, while those less used receive lower doses.[33]

For tailored treatment of glabellar lines, the experts recommended a customized treatment plan, in which patients are injected according to their individual needs in terms of dosage, targeted muscles, and injection points (Figure 7). A study that compared patients treated for glabellar lines with this approach versus a standard 5-point injection technique showed that the tailored treatment was associated with significantly higher efficacy, especially for patients presenting with asymmetric lines.[36] Following assessment of the pattern and distribution of the glabellar lines under maximum contraction, the patient may receive from 3 to 12 points of injection with Inco. These points will vary for each patient, depending on the dominance of the procerus, corrugators, orbicularis oculi, depressor supercilii, or frontalis in each contraction pattern.

4.5.5 | Ptosis management

To reduce the risk of lid ptosis when treating glabellar frown lines, experts advised placing the thumb of the noninjecting hand firmly under the superior orbital rim to avoid any diffusion of neurotoxin into the levator palpebral muscle. When injecting, the needle should be oriented upwards and towards the median line. Injections in the vicinity of the levator palpebral muscle and the cranial part of the orbicularis oculi muscle should be avoided. Injections into the corrugator should be performed at least 1 cm above the orbital rim. If patients have experienced eyelid ptosis following previous injections, the dose of Inco should be reduced. Lateral injection of the corrugator must stay superficial.

Should lid ptosis occur it can be treated with injection of botulinum toxin in the orbicularis oculi muscle as described by Sarfati et al.[37] The injection point is in the pretarsal section of the orbicularis

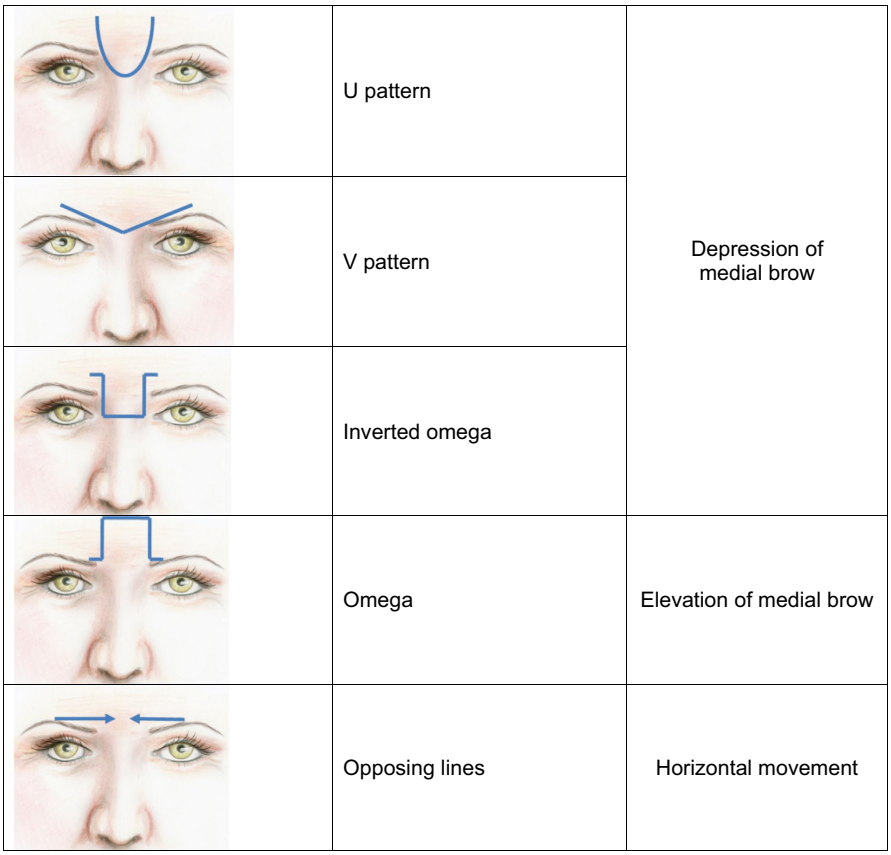


FIGURE 5 Glabellar frown line contraction patterns vary according to the direction of muscle contraction and interaction with adjacent muscles.[33]

TABLE 2 Injection protocol for glabellar frown lines

Cadaver dissection. Left corrugator muscle injection (lateral skin insertion)	
Injection sites	5 injection sites in total for 2 corrugators and the procerus muscles
Corrugator supercilii	2 injection sites in each corrugator (1 medial, 1 lateral)
Procerus	1 injection site (central)
Dosage	4–5 U Inco at each site
Total	15–20 U (women), 20–30 U (men), dependent on muscle strength. Minimal dose is 12 U
Per injection site	3–4 U (women), 4–6 U (men)
Injection technique	
Depth	Corrugators are injected deep medially (at origin oF bone) and superficial laterally (at skin insertion). Procerus is injected superficially





FIGURE 6 Standard injection protocol for glabellar frown lines.

oculi, at the mid-point where the tarsus is the highest (Figures 8–10). A toxin dose of 4 to 8 U is injected just above the eyelashes; the product is hyper-concentrated to avoid diffusion and is diluted with 0.5 mL of NaCl (50 UI botulinum toxin vial). 1 U of Abo can be used on the same site.

4.5.6 | Management of persistent glabellar lines

Persistent lines remaining after muscle relaxation may be treated with superficial injection of a hyaluronic acid dermal filler injected with a cannula.

4.6 | Crow's feet lines

4.6.1 | Relevant anatomy

Crow's feet are caused by the activity of the lateral orbicularis oculi muscle fibers, a sphincter-like muscle whose purpose is to constrict the eyes and close the eyelids (Figure 11). These lines vary in their pattern and severity according to the size and strength of the muscle. They are one of the earliest lines to appear, particularly if there is decreased skin elasticity as a result of smoking or excess UV exposure.[38] Application of BoNTA to the vertically-oriented lateral portion of the orbicularis oculi muscle prevents or softens the appearance of crow's feet lines. These same vertical fibers also pull the lateral portion of the eyebrow down, and treatment in this area can also be used to raise the tail of the eyebrow, particularly if the fibers of the frontalis extend laterally.

4.6.2 | Patient evaluation

To observe the muscle fibers of the orbicularis oculi the patient should be asked to close their eyes very tightly. The associated crow's feet line patterns and dynamic eyebrow position should be determined at rest and maximal smile. In an evaluation of 100 live subjects, four crow's feet line patterns were observed: full-fan pattern, lower-fan pattern (wrinkling in both the lower lid and upper cheek area), central-fan pattern (severe wrinkles only in the skin immediately surrounding the lateral canthus), and upper-fan pattern (wrinkles only in the upper eyelid skin down to the lateral canthus).[39] The presence of different crow's feet patterns suggests that tailored injection points and dosing are required to optimize treatment success and ensure patient satisfaction.

4.6.3 | Standard injection protocol

The recommended dose is 12 U Inco per side, i.e., 0.1 mL (4 U) at each of the three standard injection points (Table 3; Figure 12). Using Abo the suggested units can be upped to 30 U per side, but more often fewer units are enough. The first injection should be approximately 1 cm lateral to the orbital rim to avoid unwanted treatment of the muscles of ocular motion, which are inside the orbital rim, and side effects such as diplopia. Two further injections are placed approximately 1 cm above and below the first injection point, along an arc 1 cm from the bony rim (Figure 12). The most important injection point is located 1 cm lateral to the lateral canthus.

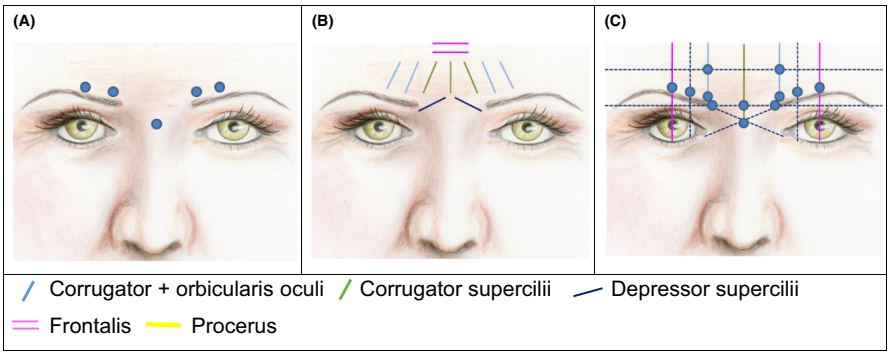


FIGURE 7 Tailored injection protocol for glabellar frown lines. (A) Standard 5-point glabellar technique, focused on the treatment of the procerus and corrugators. (B) Schematic assessment of the glabellar muscles involved in the contraction, according to the pattern of line formation in the One21 technique. (C) Schematic mapping of the One21 technique for treating the procerus, corrugators, orbicularis oculi, depressor supercilii, and frontalis.[36]

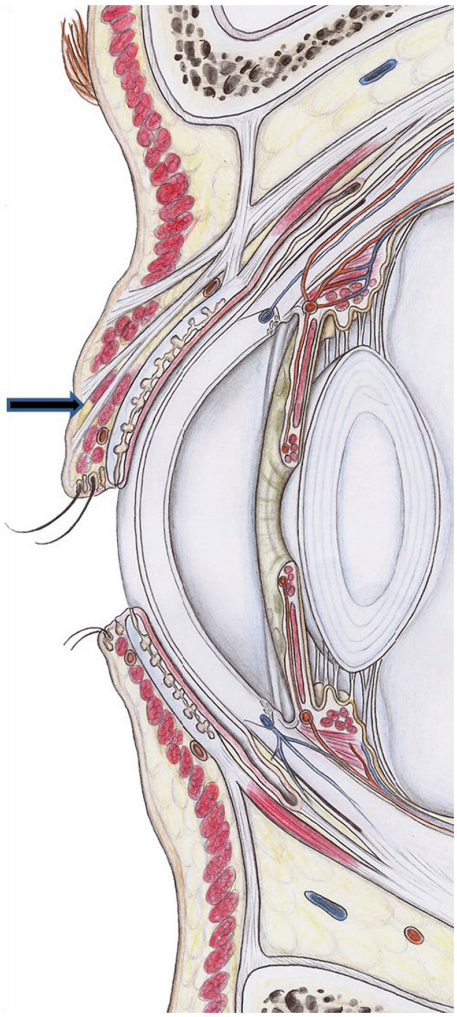


FIGURE 8 Sagittal anatomic schema to show the direction for injection in the orbicularis oculi muscle.

4.6.4 | Tailored injection protocol

The experts noted that the number of injection sites can be varied depending on the pattern and depth of the lines. In many patients, two injection sites per side (4 U each) are often enough. In patients with very wide orbicularis oculi, two rows of injection points may be required. If the physician notices descent of the tail of the eyebrow during animation, an injection into the upper orbicularis fibers may result in a raised lateral brow.

4.6.5 | Management of lower eyelid wrinkles

The experts noted that this is an indication reserved for experienced injectors. The inferior portion of the orbicularis oculi is treated very superficially with a maximum of three injection points and no more than 1 U per point. We must never forget that the orbicularis oculi is essential to maintain the lower eyelid tarse against the eyeball. Injections must respect the first centimeter along the orbital rim to avoid ectropion.

4.6.6 | Avoidance of side effects

The periorbital skin is thin and the orbicularis oculi is immediately below. All injections are performed superficially with the needle inserted almost parallel to the skin because deep injections may result in the diffusion of toxin to the zygomaticus major and/or oculomotor muscles and the risk of lip ptosis and diplopia, respectively.

The experts warned that if BoNTA injections are placed too low below the lateral canthus, the appearance of malar bags can be worsened in patients with existing malar bags. Affected patients should be informed about this risk and if the injection is carried out, the lowest injection point should be reduced to 2 U.

When injecting in this area, care should be taken to avoid venous injuries, which are very visible in the thin, transparent skin of the eye area. These can be avoided by injecting very superficially and stretching the skin tightly before each injection to visualize any vessels.

4.7 | Brow lift

A number of muscles influence brow position. The frontalis is the only brow elevator. The corrugator supercilii and the procerus depress the medial brow, and the orbicularis oculi depresses the lateral

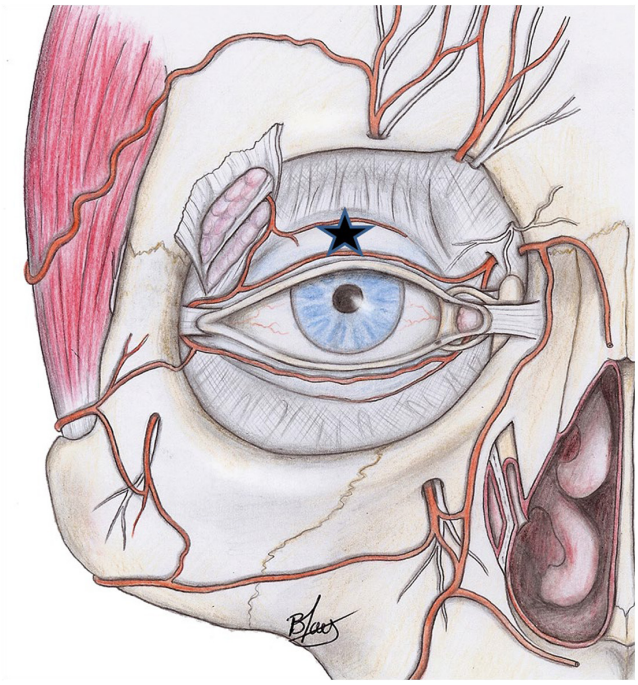


FIGURE 9 Frontal anatomic schema to show the exact location for the injection.

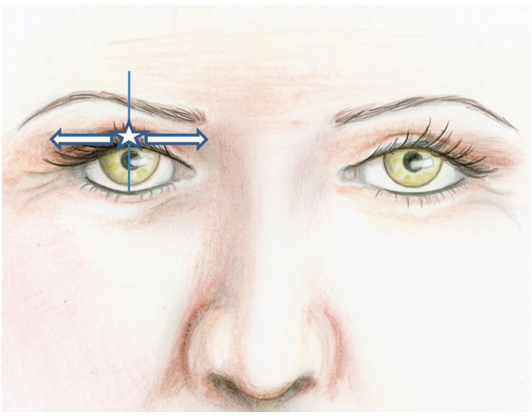


FIGURE 10 The landmark is the pupillary line.

brow. BoNTA provides injectors with a tool to adjust the interaction between the upward pull of the frontalis and the brow depressors, to achieve variations in brow shape. For example, injecting into the tail of the eyebrow (where the lateral orbicularis oculi pulls the eyebrow down) will create a small lateral brow lift. To arch and laterally raise the eyebrow, injections can be performed medial to the mid-pupillary line. By relaxing this area, the lateral frontalis will have a stronger resting tone, pulling the lateral brow up. To flatten the eyebrow and create a more male brow orientation, the point intersecting with the mid-pupillary line should be injected. To shape the brows, BoNTA injections should take place along the lowest horizontal forehead line, using low dosages (0.5–1 U Inco).[11]



FIGURE 11 Crow's feet are caused by the activity of the lateral orbicularis oculi muscle fibers.

4.8 | Follow-up and prevention of complications

Numerous studies have established the efficacy and safety of Inco for the treatment of upper facial lines.[27,28,40–42] Furthermore, a retrospective survey of European physicians using Inco for aesthetic indications (of whom 20% were from France) reported very high (>95%) physician and patient satisfaction across all approved upper face indications.[43] To maximize patient satisfaction, it is important to inform the patient of what to expect after injection and how the treatment effects progress. Many patients will recognize treatment benefits in 3–5 days, but it should be explained to patients that it may take up to 14 days for the full effect to develop. Discussions should also include the possibility of an imperfect result or asymmetry, which can be rectified if necessary with a follow-up visit 2 weeks after the initial injection. A follow-up visit at this time also allows injectors to observe how individual muscles reacted, and to determine whether higher or lower doses are required, which can improve outcomes at future visits. Managing patient expectations of BoNTA treatments and scheduling post-treatment appointments for evaluation of the initial procedure are known to increase patient satisfaction and retention rates.[44]

Detailed post-treatment advice is documented in the Inco instructions for use.[26] Patients should be advised not to massage or apply pressure to the treated area, and in particular to avoid massage towards the interior of the orbit for the first 24 h after injection. Further precautions in the first 24 h include avoidance of rigorous exercise and excessive sun or heat exposure. There are no contraindications to air travel on the day of BoNTA treatment, although experts suggested that delaying travel for 24 h would be preferable. All agreed that normal life could resume after 24–48 h.

5 | SUMMARY AND CONCLUSIONS

This consensus is intended for both novice and more experienced injectors with the aim of optimizing patient outcomes following

TABLE 3 Injection protocol for crow's feet lines

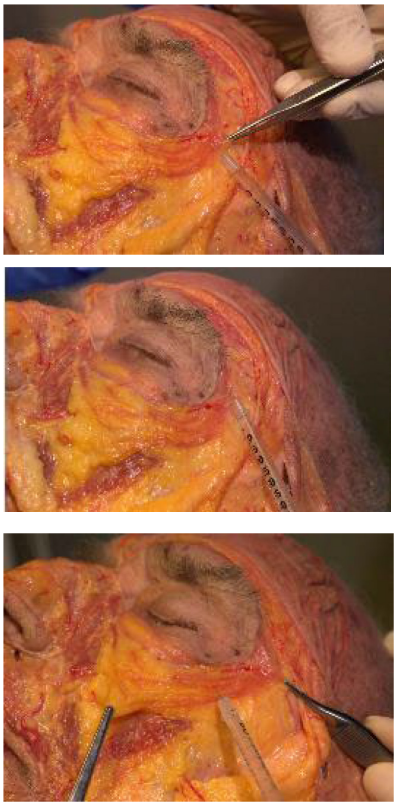
Cadaver dissection. Left orbicularis oculi muscle injection. 3 points of injection	
	
Injection sites	3 injection sites (additional row lateral to the first may be added in patients with wide orbicularis oculi)
Dosage	
Total	12–24 U (women), 18–36 U (men), dependent on muscle strength. Minimal dose is 12 U
Per injection site	2–4 U (women), 3–6 U (men), 4–10 Abo
Injection technique	
Depth	Superficial
Angle	Almost parallel
Direction	Medial to lateral direction to keep away from the orbit



FIGURE 12 Standard injection protocol for crow's feet lines.

treatment with Inco for upper facial lines. Although BoNTAs have been approved for aesthetic use for nearly 20 years, injection techniques continue to evolve. There is now less reliance on standardized templates for injection sites and doses with a strong move towards tailored treatment, taking into account individual facial anatomy, patterns and strength of muscle activity, and patient preferences. This requires injectors to assess patients' muscles while at rest and during animation, and to have a thorough understanding of muscular anatomy. Knowledge of muscle origin and skin insertion points can help the injector decide which depth or plane should be injected for the best outcome, and how interactions between muscle groups influence the position of facial features such as the brow. Facial anatomy differs widely from person to person, and variations in muscle configuration create different wrinkling patterns of varying severity. The injector should be aware of these variations and within each muscle, target zones of high strength with larger doses or more injections, and zones of lower strength with smaller doses or none at all. This requires injections to be directed and precise, as temporary disfigurement or functional impairment can occur if BoNTA inadvertently diffuses into the adjacent muscle that has an opposing effect.

Younger patients are increasingly seeking BoNTA treatment as a preventive measure to avoid static lines developing while retaining natural facial expressions. This requires a precise injection technique with small volume doses at varying tissue depths and is a further move away from the standard injection protocols.

Finally, for a more natural-looking effect and smoother transition between treated areas, it is now common place for physicians to perform combined treatment of the different upper facial areas targeting all three types of line during the same treatment session instead of isolation.

AUTHOR CONTRIBUTIONS

FB wrote the paper, with individual sections provided by IC, FBe, JL, ER, JP, JH, HC, MD, OG, DO, HS, ES, PK, CW, and AR. All authors reviewed and commented on the manuscript. FB coordinated the

reviews and input the comments. All authors read and approved the final version of the manuscript.

DATA AVAILABILITY STATEMENT

The referenced data are available in the public domain via the PubMed site.

ETHICS STATEMENT

The authors are accountable for all aspects of the work and in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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REFERENCES

1. International Society of Aesthetic Plastic Surgery (ISAPS). ISAPS international survey on aesthetic/cosmetic procedures performed in 2019. Available from: https://www.isaps.org/wp-content/uploads/2020/12/Global-Survey-2019.pdf. Accessed Nov 14, 2022.
2. Samizadeh S, De Boulle K. Botulinum neurotoxin formulations: overcoming the confusion. *Clin Cosmet Investig Dermatol*. 2018;11:273-287.
3. Nestor MS, Arnold D, Fischer DL. The mechanisms of action and use of botulinum neurotoxin type a in aesthetics: key clinical postulates II. *J Cosmet Dermatol*. 2020;19(11):2785-2804.
4. Kerscher M, Roll S, Becker A, Wigger-Alberti W. Comparison of the spread of three botulinum toxin type a preparations. *Arch Dermatol Res*. 2012;304(2):155-161.
5. Grein S, Mander GJ, Fink K. Stability of botulinum neurotoxin type a, devoid of complexing proteins. *Botulinum J*. 2011;2(1):49-58.
6. Kerscher M, Wanitphakdeedecha R, Trindade de Almeida A, Maas C, Frevert J. Incobotulinumtoxin A: a highly purified and precisely manufactured botulinum neurotoxin type a. *J Drugs Dermatol*. 2019;18(1):52-57.
7. Carr WW, Jain N, Sublett JW. Immunogenicity of botulinum toxin formulations: potential therapeutic implications. *Adv Ther*. 2021;38(10):5046-5064.
8. Yutskovskaya Y, Gubanova E, Khrustaleva I, et al. Incobotulinumtoxin A in aesthetics: Russian multidisciplinary expert consensus recommendations. *Clin Cosmet Investig Dermatol*. 2015;8:297-306.
9. Freeman M, Sayegh F, Sarosi A, et al. Millennials are interested in botulinum toxin injections for prevention of facial rhytids. *Face*. 2021;2(1):94-98.
10. Anido J, Arenas D, Arruabarrena C, et al. Tailored botulinum toxin type a injections in aesthetic medicine: consensus panel recommendations for treating the forehead based on individual facial anatomy and muscle tone. *Clin Cosmet Investig Dermatol*. 2017;10:413-421.
11. de Sanctis PC. One21: a novel, customizable injection protocol for treatment of the forehead with Incobotulinumtoxin A. *Clin Cosmet Investig Dermatol*. 2020;13:127-136.
12. Galadari H, Galadari I, Smit R, Prygova I, Redaelli A. Use of abobotulinumtoxin A for cosmetic treatments in the neck, and middle and lower areas of the face: a systematic review. *Toxins (Basel)*. 2021;13(2):169.
13. Sundaram H, Signorini M, Liew S, et al. Global aesthetics consensus: botulinum toxin type a – evidence-based review, emerging concepts, and consensus recommendations for aesthetic use, including updates on complications. *Plast Reconstr Surg*. 2016;137(3):518e-529e.
14. de Maio M, Swift A, Signorini M, Fagien S. Aesthetic leaders in facial aesthetics consensus committee. Facial assessment and injection guide for botulinum toxin and injectable hyaluronic acid fillers: focus on the upper face. *Plast Reconstr Surg*. 2017;140(2):265e-276e.
15. Kaminer MS, Cox SE, Fagien S, Kaufman J, Lupo MP, Shamban A. Re-examining the optimal use of neuromodulators and the changing landscape: a consensus panel update. *J Drugs Dermatol*. 2020;19(4):s5-s15.
16. Fabi SG, Carruthers J, Joseph J, et al. High-dose neuromodulators: a roundtable on making sense of the data in real-world clinical practice. *Aesthet Surg J Open Forum*. 2021;3(4):ojab036.
17. Yoelin SG, Kwolek MS, Block JE. Commentary on botulinum toxin injections: a call for improved accuracy. *J Clin Aesthet Dermatol*. 2020;13(8):13-14.
18. Costin BR, Wyszynski PJ, Rubinstein TJ, et al. Frontalis muscle asymmetry and lateral landmarks. *Ophthalmic Plast Reconstr Surg*. 2016;32(1):65-68.
19. Jia Z, Lu H, Yang X, et al. Adverse events of botulinum toxin type a in facial rejuvenation: a systematic review and meta-analysis. *Aesthetic Plast Surg*. 2016;40(5):769-777.
20. Carey WD. Incorrect reconstitution of incobotulinumtoxin A leads to loss of neurotoxin. *J Drugs Dermatol*. 2014;13(6):735-738.
21. Niamtu J III. Neurotoxin waste from drawing product through the vial stopper. *J Clin Aesthet Dermatol*. 2014;7(6):33-37.
22. Jost WH, Benecke R, Hauschke D, et al. Clinical and pharmacological properties of incobotulinumtoxin A and its use in neurological disorders. *Drug des Devel Ther*. 2015;9:1913-1926.
23. Evidente VG, Fernandez HH, LeDoux MS, et al. A randomized, double-blind study of repeated incobotulinumtoxin A (Xeomin(®)) in cervical dystonia. *J Neural Transm (Vienna)*. 2013;120(12):1699-1707.
24. Truong DD, Gollomp SM, Jankovic J, et al. Sustained efficacy and safety of repeated incobotulinumtoxin A (Xeomin(®)) injections in blepharospasm. *J Neural Transm (Vienna)*. 2013;120(9):1345-1353.
25. Wissel J. Towards flexible and tailored botulinum neurotoxin dosing regimens for focal dystonia and spasticity – insights from recent studies. *Toxicon*. 2018;147:100-106.
26. Bocouture 100 units—summary of product characteristics (SmPC)—(emc). https://www.medicines.org.uk/emc/product/7418/smpc#gref. Accessed September 4, 2022
27. Coleman WP 3rd, Sattler G, Weissenberger P, Hast MA, Hanschmann A. Safety of Incobotulinumtoxin A in the treatment of facial lines: results from a pooled analysis of randomized, prospective, controlled clinical studies. *Dermatol Surg*. 2017;43(Suppl 3):S293-S303.
28. Fischer T, Sattler G, Prager W, et al. Safety, tolerability, and efficacy of repeat-dose injections of Incobotulinumtoxin A in the treatment

of upper facial lines: results from a prospective, open-label, phase III study. *J Drugs Dermatol*. 2020;19(5):461-469.
29. Abramo AC, Do Amaral TP, Lessio BP, De Lima GA. Anatomy of forehead, glabellar, nasal and orbital muscles, and their correlation with distinctive patterns of skin lines on the upper third of the face: reviewing concepts. *Aesthetic Plast Surg*. 2016;40(6):962-971.
30. de Sanctis PC, Bussamara Pinheiro MV, Figueiredo V, Guerra R. Aesthetic one21 technique for injecting Incobotulinumtoxin A into the forehead: initial experience with 86 patients. *J Drugs Dermatol*. 2021a;20(1):31-37.
31. Hwang K, Lee JH, Lim HJ. Anatomy of the corrugator muscle. *J Craniofac Surg*. 2017;28(2):524-527.
32. Lee HJ, Lee KW, Tansatit T, Kim HJ. Three-dimensional territory and depth of the corrugator supercilii: application to botulinum neurotoxin injection. *Clin Anat*. 2020;33:795-803.
33. de Almeida AR, da Costa Marques ER, Banegas R, Kadunc BV. Glabellar contraction patterns: a tool to optimize botulinum toxin treatment. *Dermatol Surg*. 2012;38(9):1506-1515.
34. Redaelli A, Saromytskaya A, Rowland PC. International expert consensus on the use of AboBotulinum toxin a (AboTA) for facial rejuvenation and primary hyperhidrosis. *CosMed*. 2017;3(2):70-80.
35. Prager W. Differential characteristics of incobotulinumtoxin A and its use in the management of glabellar frown lines. *Clin Pharmacol*. 2013;5:39-52.
36. de Sanctis PC, Pinheiro MVB, Ventura Ferreira K, de Barros J, Nunes G, Miot HA. The One21 technique: an individualized treatment for glabellar lines based on clinical and anatomical landmarks. *Clin Cosmet Investig Dermatol*. 2021b;14:97-105.
37. Sarfati E, Esposito-Valat F, Nefzaoui C. Ptosis traités par injection de toxine botulique: une série de 31 patients. *J Méd Chirurgie Dermatol*. 2020;199-217.
38. Lee JY, Kim YK, Seo JY, et al. Loss of elastic fibers causes skin wrinkles in sun-damaged human skin. *J Dermatol Sci*. 2008;50(2):99-107.
39. Kane MA, Cox SE, Jones D, Lei X, Gallagher CJ. Heterogeneity of crow's feet line patterns in clinical trial subjects. *Dermatol Surg*. 2015;41(4):447-456.
40. Kerscher M, Rzany B, Prager W, Turnbull C, Trevidic P, Inglefield C. Efficacy and safety of Incobotulinumtoxin A in the treatment of upper facial lines: results from a randomized, double-blind, placebo-controlled, phase III study. *Dermatol Surg*. 2015;41(10):1149-1157.
41. Prager W, Nogueira Teixeira D, Leventhal PS. Incobotulinumtoxin A for aesthetic indications: a systematic review of prospective comparative trials. *Dermatol Surg*. 2017;43(7):959-966.
42. Trevidic P, Connolly SA, Biwer B, et al. Incobotulinumtoxin A is an effective and well-tolerated treatment for upper facial lines: results from an open-label extension period of a phase III study. *Dermatol Surg*. 2017;43(Suppl 3):S285-S292.
43. Pavicic T, Prager W, Klöppel M, Ravichandran S, Galatoire O. Incobotulinumtoxin A use in aesthetic indications in daily practice: a European multicenter, noninterventional, retrospective study. *Clin Cosmet Investig Dermatol*. 2015;8:135-142.
44. White L, Tanzi EL, Alster TS. Improving patient retention after botulinum toxin type a treatment. *Dermatol Surg*. 2006;32(2):212-215

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